

7CS353 Machine Learning Lab — Assignment 6

Multiple Linear Regression: Model Fitting, Diagnostics, and Refinement

Class, Semester	Third Year B. Tech. (CSE), Sem V — AY 2025-26
Linked courses	7CS353 (Lab) and 7CS302 Module III (Regression)
Builds on	Assignment 5 — SLR, Normal Equation, error metrics
Structure	Part A: Theoretical concepts (not graded) Part B: MLR analysis (15 marks — CO2 + CO3)
Duration	1 lab session (2 hrs), submit within 1 week

0. Dataset — data_MLR.csv

The dataset contains 1000 observations with 11 predictor variables (x_1 to x_{11}) and one continuous response variable (y). There are no missing values. The predictors have not been pre-screened — some are highly correlated with each other and not all contribute meaningfully to predicting y . Students are required to load the file, explore it, and systematically build and refine an MLR model following the workflow in A.6.

PART A — Theoretical Concepts

Read before starting Part B. Not graded.

A.1 Model and Coefficient Interpretation

MLR extends SLR by allowing multiple predictors. The population model for $p-1$ predictors is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{p-1} x_{p-1} + \epsilon$$

Interpretation of β_j . Expected change in y per unit increase in x_j , when all other predictors are held constant. This "all else equal" clause is what distinguishes MLR coefficients from simple pairwise correlations — the coefficient absorbs the shared influence of the other features in the model.

p-value of a coefficient. Each fitted coefficient $\hat{\beta}_j$ is tested with $H_0: \beta_j = 0$ — the predictor adds no information given the others currently in the model. A p-value below 0.05 provides evidence that the predictor is statistically significant in the presence of the other features. Note that significance can change as features are added or removed.

A.2 Matrix Form and Normal Equation

Stacking all n observations, the model is written in matrix form as $y = X\beta + \epsilon$, where X is the $n \times p$ design matrix with a leading column of 1s for the intercept. The OLS solution that minimises the sum of squared errors is:

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Hat matrix. The fitted values are $\hat{y} = Xb = X(X^T X)^{-1} X^T y = Hy$, where $H = X(X^T X)^{-1} X^T$ is the hat matrix. It is called the hat matrix because it puts the hat on y — it maps observed responses to fitted values. The diagonal elements h_{ii} play a central role in diagnosing leverage (see A.5).

A.3 Adjusted R^2

The limitation of R^2 in MLR. Adding any predictor — even a column of random noise — can only increase (or leave unchanged) R^2 . A model with 11 predictors will always appear to fit at least as well as a model with 3,

regardless of whether the extra predictors carry real information. This makes R^2 a poor basis for comparing models of different sizes.

Quantity	Formula
R^2 <i>Increases monotonically with added predictors</i>	$R^2 = 1 - \text{SSE} / \text{SST}$
Adjusted R^2 <i>Penalises for p; decreases when a predictor adds less than it costs — use this for model comparison in MLR</i>	$R^2_{\text{adj}} = 1 - (\text{SSE}/(n-p)) / (\text{SST}/(n-1))$

n = number of observations, p = number of parameters including the intercept. When comparing two models, always prefer the higher Adjusted R^2 . If adding a predictor causes Adjusted R^2 to drop, the predictor is not worth keeping.

A.4 Multicollinearity and VIF

What multicollinearity is. Multicollinearity occurs when two or more predictors are nearly linearly dependent on each other. When this happens, $X^T X$ becomes nearly singular — its inverse is numerically unstable. The practical consequences are: (1) fitted coefficients $\hat{\beta}_j$ have very large standard errors, so they fluctuate wildly across different samples; (2) the coefficient of x_k is confounded with the coefficient of the features it is correlated with, making individual interpretation unreliable; (3) p-values become unreliable — a feature may appear insignificant simply because its information is shared with another feature already in the model.

Variance Inflation Factor (VIF). For each predictor x_k , temporarily treat x_k as the response and regress it on all remaining predictors. Compute R^2_k — the proportion of variance in x_k explained by the other predictors. The VIF converts this into a measure of collinearity:

$$\text{VIF}_k = 1 / (1 - R^2_k)$$

When R^2_k is close to 1 (x_k is almost perfectly predictable from the others), VIF_k becomes very large. The denominator $1 - R^2_k$ is the fraction of variance in x_k that is unique to it — the smaller this is, the more the variance of $\hat{\beta}_k$ is inflated relative to a scenario with no collinearity.

VIF range	Action
$\text{VIF} = 1$	No correlation with other predictors — keep the feature.
$1 < \text{VIF} \leq 4$	Mild correlation — generally acceptable.
$4 < \text{VIF} \leq 10$	Moderate concern — use cross-validation to decide whether to retain.
$\text{VIF} > 10$	Serious multicollinearity — drop the feature or combine with the correlated predictor.

A.5 Outliers, Leverage, and Influence

Three related but distinct concepts characterise problematic observations in a regression dataset. Understanding the difference between them is essential because each requires a different diagnostic and a different response.

Outliers

An outlier in regression is a data point whose response value y is unusual given its predictor values x — that is, the model produces a large residual for that observation. Outliers may arise from data entry errors, measurement failures, or genuinely rare events.

Detection — internally standardized residuals. The raw residual $e_i = y_i - \hat{y}_i$ is not on a common scale because the variance of each residual depends on leverage. Dividing by the estimated standard deviation gives the internally standardized residual:

$$r_i = e_i / (s \sqrt{1 - h_{ii}})$$

where $s = \sqrt{SSE/(n-p)}$ is the residual standard error

Flag an observation as an outlier if $|r_i| > 3$. The threshold of 3 corresponds roughly to a 99.7% interval under normality — values beyond this are expected to occur very rarely by chance in a well-fitting model.

Impact of outliers. Outliers inflate MSE and SSE, which directly reduces R^2 and Adjusted R^2 . They can also pull the fitted line toward them, distorting the coefficients. However, not every outlier is influential — an outlier with a predictor value close to the mean of x will have little leverage and may not materially affect the fit.

High Leverage

A high-leverage observation has an extreme or unusual combination of predictor values — it lies far from the centre of the X space, even if its response value y is perfectly consistent with the model. Leverage depends entirely on X , not on y .

Formal definition. The leverage of observation i is the i -th diagonal element h_{ii} of the hat matrix $H = X(X^T X)^{-1} X^T$. It has two key properties:

- $h_{ii} \in [0, 1]$ for all observations, and $\sum h_{ii} = p$ (the trace of H equals the number of parameters).
- $h_{ii} = 1/n$ for all points when X is balanced (e.g., an orthogonal design). Higher h_{ii} means observation i exerts more influence on its own fitted value $\hat{y}_i$.

Detection threshold. A commonly used rule of thumb flags an observation if $h_{ii} > 2p/n$. This threshold says a point is "high leverage" if its hat value is more than twice the average leverage p/n .

High leverage is not automatically harmful. A high-leverage point whose response is consistent with the fitted model (i.e., its residual is small) will reinforce the regression line rather than distort it. The danger arises when a high-leverage point also has an unusual response value — then it can pull the entire fitted line toward itself and away from the bulk of the data.

Influential Points

An influential observation is one that substantially changes the fitted regression coefficients when it is included versus when it is excluded. Influence is the combination of high leverage and an unusual response — a point that is both extreme in X and not well-described by the model for the remaining data.

Cook's Distance. Cook's D measures the aggregate shift in all n fitted values $\hat{y}$ when observation i is deleted from the dataset and the model is refit:

$$D_i = (\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(-i)})^T (\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(-i)}) / (p \cdot \text{MSE})$$

$$\text{equivalently: } D_i = (r_i^2 / p) \cdot (h_{ii} / (1 - h_{ii}))$$

The second form is illuminating: Cook's D is the product of how much of an outlier the point is (r_i^2) and how much leverage it has ($h_{ii}/(1-h_{ii})$). A point that is a moderate outlier AND has high leverage can have a very large Cook's D even if neither alone is extreme.

Threshold. The common conservative threshold is $D_i > 4/n$. A more lenient threshold is $D_i > 1$. In practice, Cook's D is better interpreted graphically — a stem plot or index plot reveals which points stand out relative to the rest.

DFFITS. DFFITS measures the change in the predicted value for observation i specifically (not all fitted values) when it is deleted, expressed in units of standard deviations:

$$\text{DFFITS}_i = (\hat{y}_i - \hat{y}_{(i)}) / (s_{(i)} \sqrt{h_{ii}})$$

$\hat{y}_{(i)}$ is the predicted value at x_i from the model fitted without observation i , and $s_{(i)}$ is the residual standard error from the model without i . Flag if $|\text{DFFITS}_i| > 2\sqrt{p/n}$.

What to do with influential points. Identifying an influential observation is not sufficient reason to delete it — deletion requires justification. Check whether the point is a data error (then correct or remove it). If it is a genuine observation, consider whether the model should be expanded to accommodate that region of X . If removal substantially improves Adjusted R^2 and the point cannot be explained, document the decision clearly and report results both with and without the point.

A.6 MLR Workflow

Follow these steps in order when fitting an MLR model to new data. Each step may require iterating back if the output changes what was assumed in a prior step.

Step	Action
1	Split data into training (80%) and test (20%) sets. Normalise features using StandardScaler fitted on training data only — never fit the scaler on test data.
2	Fit the full model on all predictors using statsmodels OLS. Record Adjusted R^2 , overall F-test p-value, and individual coefficient p-values.
3	Compute VIF for every predictor. Drop features with $\text{VIF} > 10$ one at a time (highest VIF first). Refit and recompute VIF after each drop. Repeat until no VIF exceeds 10.
4	On the current model, check Cook's Distance and DFFITS. Plot Cook's Distance with a threshold line at $4/n$. For any flagged observations, refit without them and check whether Adjusted R^2 improves. Document every point that is removed.
5	From the current feature set, identify predictors with p-value > 0.05 . Drop them one at a time, starting with the highest p-value. Refit after each removal. Stop when all remaining predictors are significant or Adjusted R^2 begins to decrease.
6	On the final model, draw a residuals vs fitted plot and a Q-Q plot of residuals. Assess whether the LINE assumptions (Linearity, Independence, Normality, Equal variance) are satisfied. If patterns suggest non-linearity, consider transformations of X or y .
7	Report final model coefficients, Adjusted R^2 , MAE, RMSE, and R^2 on the held-out test set. Interpret each remaining coefficient and compare training Adjusted R^2 with test R^2 to assess generalisation.

PART B — MLR Analysis (15 marks)

Use Part A as your reference. Each step in its own cell with a one-line markdown comment above it. All plots must have a title and labelled axes.

Perform a complete Multiple Linear Regression analysis on data_MLR.csv by implementing the 7-step workflow described in A.6.

For each step, write your code in its own notebook cell and document every model decision — which feature was dropped and why, which points were removed and why, and why you stopped feature refinement where you did — in a markdown cell immediately below the code. The quality of your written justifications is assessed equally with the correctness of the code.

Your analysis must produce, at minimum:

- A VIF table at each iteration showing the feature dropped and the Adjusted R^2 after dropping it.
- A Cook's Distance plot with threshold line; a clear statement of which observations, if any, were removed.
- A stepwise refinement log showing Adjusted R^2 after each feature is dropped.
- Residuals vs fitted and Q-Q plots for the final model.
- A test-set evaluation table: MAE, RMSE, R^2 , and a comparison with training Adjusted R^2 .
- A final markdown summary: the retained features, the sign and interpretation of each coefficient, and your assessment of model quality.

Deliverables

- One notebook PRN_Assignment6.ipynb — full pipeline from data loading to test evaluation.
- All plots with titles and labelled axes; unlabelled plots count as incomplete.
- Markdown justification cell after every decision step.
- Submission as per earlier guidelines.
- Viva at the start of the next session — be ready to justify every decision.

References

1. Penn State STAT 501 — Regression Methods, Lesson 11: Influential Points. <https://online.stat.psu.edu/stat501/Lesson11> (Primary reference for outliers, leverage, Cook's Distance, and DFFITS with worked examples.)
2. Rick Wicklin, SAS IML Blog — Influential Observations in a Linear Regression Model: The DFFITS and Cook's D Statistics. <https://blogs.sas.com/content/iml/2019/06/19/influence-regression-dffits-cooksd.html> (Detailed derivation of both statistics and their relationship.)
3. R. W. Nahhas — Introduction to Regression Methods for Public Health Using R, Chapter 5.23: Influential Observations. <https://www.bookdown.org/rwnahhas/RMPH/mlr-influence.html> (Practical interpretation and sensitivity analysis guidance.)

Assessment Rubric (15 marks — Part B, toward LA2)

Part B is assessed against two Course Outcomes. CO2 covers execution of the MLR analysis pipeline; CO3 covers the quality of decisions and justifications throughout.

CO	Weight	Rubric statement	Very good	Good	Average	Needs improvement	Not done
CO2	10 marks	Execute the complete MLR workflow: fit the full model, diagnose multicollinearity (VIF), detect influential observations (Cook's D, DFFITS), refine features stepwise, conduct residual	All 7 workflow steps complete; diagnostics correct; plots labelled; metrics computed accurately.	5–6 steps complete; minor errors in diagnostics or plots; metrics present.	3–4 steps complete; significant gaps in diagnostics or visualisation.	1–2 steps complete; major parts of the pipeline missing.	Not submitted or no meaningful analysis.

		analysis, and evaluate on test data.					
CO3	5 marks	Justify every model decision: explain which features to drop and why, evaluate whether influential observations should be removed, compare model variants, and assess whether the final model generalises well.	Every decision justified with reference to diagnostic values; model comparison clear; generalisation assessed with evidence.	Most decisions justified; comparison mostly clear; minor gaps.	Some justifications present but shallow or unsupported by diagnostic values.	Little or no justification; decisions made without explanation.	No written interpretation attempted.
Total		15 marks (CO2 = 10 + CO3 = 5)					